

Lab 09: Modeling for Decision Making

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Objective: In this lab, students will try to propose a solution to an open-ended problem based on their respective understanding of the given scenario.

Instructions: Below are some instructions. Please go through them carefully:

- Given below is a scenario where students need to propose a solution-based
- Use the traditional approach to name your files for submission:
 - `ROLLNUMBER_IE507_Lab09.ipynb`;
 - `ROLLNUMBER_IE507_Lab09_Report.pdf`

Question 1: Efficient Tour Planning- A Heuristic Approach

Consider that **S1** and **S2** are two cities miles apart as hometown and workplace (are known as source nodes, **S**). A list of cities of attractions for your itinerary is given as a set **C** (representing the cities you plan to visit along with your parents). Further, these cities of attractions are distributed over a geographical region, and direct connectivity may/may not be possible from one tourist attraction to the other. Both **S1** and **S2** belong to metropolitan regions, and a direct route is possible from **S** to **C**. But you are constrained by choice to select a meeting point from a set **M**, which intersects on **S1 – S2** route. All cities in **C** are well spread and may be connected from **M** via. some distance and trains respectively. Below is a graph in Fig. 1 for the network connecting various cities in **M** and **C**. Note that a dashed-blue vertex is between the edges of elements of **C** with a meeting point in **M**. A black vertex represents the route link between a meeting point in set **M** and a source city from **S**. Below is a sample layout for a graph with two source nodes, one meeting points, and one city of attractions. Further, we provide you with some datasets to describe the characteristics of the below network.

Data Set ($\forall i, j \in S \cup C \cup M$)

- **S** = {**S1**, **S2**} represents source nodes
- **C** = {**C1**, **C2**, ..., **Cp**} represents nodes for cities of attraction where $p = 1$
- **M** = {**M1**, **M2**, ..., **Mq**} represents nodes for possible meeting points where $q = 1$
- **N** = $[n]_{ij}$ denotes the *adjacency matrix* representing the possible route between node i to node j .

	S1	S2	M1	C1
S1	0	0	1	0
S2	0	0	1	0
M1	1	1	0	1
C1	0	0	1	0

- **D** = $[d]_{ij}$ denotes the *distance matrix* representing the distance (in kms) between node i to node j .

	S1	S2	M1	C1
S1	0	0	1508	0
S2	0	0	807	0
M1	1508	807	0	167
C1	0	0	167	0

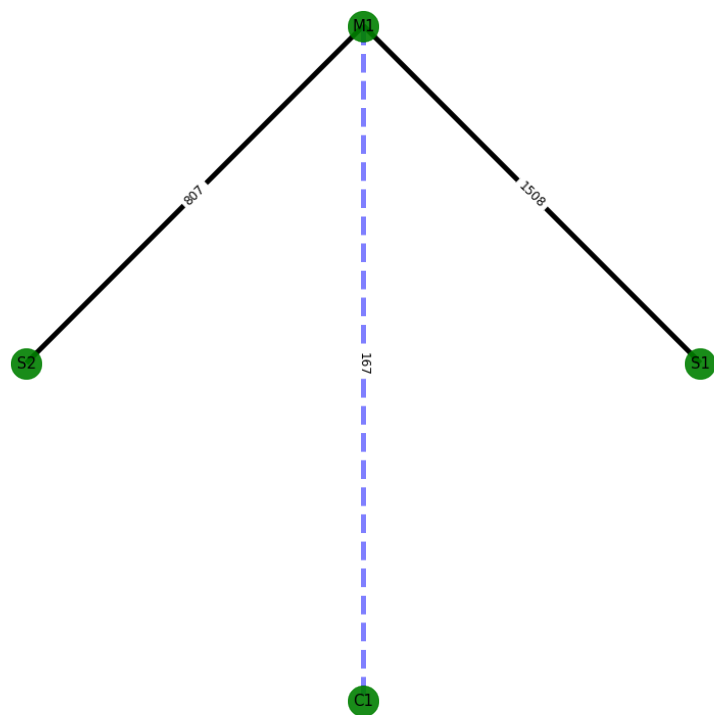


Figure 1: A layout for network with two source cities (**S1**,**S2**), one meeting point **M1** and one city of attraction **C1**

- $\mathbf{F} = [\mathbf{f}]_{ij}$ denotes the *frequency matrix* representing the daily frequency of trains between node i to

		S1	S2	M1	C1
node j .	S1	0	0	8	0
	S2	0	0	5	0
	M1	8	5	0	6
	C1	0	0	6	0

- $\mathbf{TT} = [\mathbf{tt}_k]_{ij}$ denotes the *travel time matrix* representing the travel time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$. Note that travel time is symmetric that is travel time for node j to node i is same as node i to node j .

S1 to M1: [18:35, 21:27, 19:46, 26:25, 28:36, 20:28, 32:16, 22:10]

S2 to M1: [16:25, 19:38, 20:56, 09:13, 12:08]

M1 to C1: [03:42, 02:56, 04:15, 03:18, 05:08, 03:50]

- $\mathbf{DEP} = [\mathbf{dep}_k]_{ij}$ denotes the *departure time matrix* representing the departure time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$.

S1 to M1: [16:30, 14:40, 03:55, 01:25, 10:26, 12:25, 06:35, 20:15]

M1 to S1: [18:25, 06:20, 02:25, 05:15, 09:20, 15:15, 20:18, 22:20]

S2 to M1: [04:20, 08:35, 12:15, 18:12, 22:18]

M1 to S2: [12:25, 15:16, 18:21, 07:25, 21:20]

M1 to C1: [05:50, 10:25, 12:30, 18:15, 21:05, 16:20]

C1 to M1: [04:00, 07:15, 10:55, 18:35, 21:25, 23:50]

- $\mathbf{ARR} = [\mathbf{arr}_k]_{ij}$ denotes the *arrival time matrix* representing the arrival time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$.

NOTE: Model the scenario below using your programming skills so that your proposed solution works for large networks. For simplicity, assume that there is zero stoppage time at each node. Also, all the train timetable is in a 24-hour format.

Scenario A: Creating DataFrames

1. You are given above the travel and departure times for each type of train on each network link. Identify the Arrival Time for the above trains. Also, create a dataframe (or a suitable numpy array) using the above data for Arrival and Departure with the *row as From-station* and the *column as To-station*.
2. Execute a code to create an Arrival-Departure Board for all the nodes based on the above-given data set.

Note ¹: While creating dataframe for Arrival-Departure Board each train must be named for identification. For more details, refer to the footnote given below.

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$< \text{DepartureTime}(\text{at Origin Node}) > - < \text{OriginNode} > < \text{DestinationNode} > - < \text{ArrivalTime}(\text{at Dest. Node}) >$

A sample entry on board might look like

S1 $\rightarrow$ **M1**(05:30_S1C1.05:15): Departing **S1** @ 05:30 and reaching **M1**@01 : 30 (+1 day).

Scenario B: Scheduling Intersection

Note ²: While providing an output of a feasible itinerary, you need to use an approach similar to the elementary set theory; for more details, refer to the footnote below.

1. A **layover** in our context significantly differs from the **waiting time** at the meeting point. Layover is when both parties come separately and wait for the next leg of the journey. At the same time, *waiting time* here refers to either of the parties reaching early at the meeting point and waiting for the other party.
 - Create a plot for feasible trains arriving at the meeting point **M1** arriving from **{S1, S2}** against the possible number of feasible departures for **C1** with a minimum layover time of 1 hour. You should look for a feasible train from **M1** to **C1**, whose departure from **M1** is up to 24 hours from the time of first arrival at **M1** (this is same as saying that waiting_time+layover_time is less than 24 hours).
 - Also, report your feasible travel plan using the prescribed notation.
2. Repeat the same if the minimum layover time belongs to the set $\{2, 5\}$ hours. Describe your obtained plot based on your understanding.
3. Assume that your tour planning is now constrained by waiting time at meeting point **M1** say, waiting time not more 1.5 hours. You can continue to assume a minimum layover time of 2 hours. Plot your feasible set of trains from the source nodes **{S1, S2}** against departing trains at **M1** to **C1** with required layovers and waiting time. Describe your obtained plot based on your understanding. Also, report your feasible travel plan using the prescribed notation.
4. Repeat part 3 with a maximum waiting time of 2.5 hours.

Hint: Try to use a heatmap as a plot!!!

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$\{\{\text{Train No. for S1 to M1 train, Train No. for S2 to M1, \{\text{Train No. for all feasible train from M1 to C1}\}, \dots \{\dots\}\}\}$

In case if no feasible mode is available for a certain combination of train **S1, S2** return feasible solution as $\{\phi\}$